



Management and Cost Accounting – Costing Applications

Exercise 3.2

“Buttercup Bakery” is a local bakery known for its handcrafted breads and pastries. They are considering expanding their operations by operating a second location in a nearby town. This expansion will significantly increase their fixed costs (rent, electricity) and potentially change their variable costs (ingredients, packaging) due to increased volume.

Task 1

Draw a generic Cost-Volume-Profit (CVP) graph including the following lines: **Fixed cost line**, **variable cost line**, and **total cost line**. Clearly label the following on your graph: **Breakeven point**, **variable costs**, **fixed costs**, **operating profit area**, and **operating loss area**.

Label the axes:

X-axis: units sold.

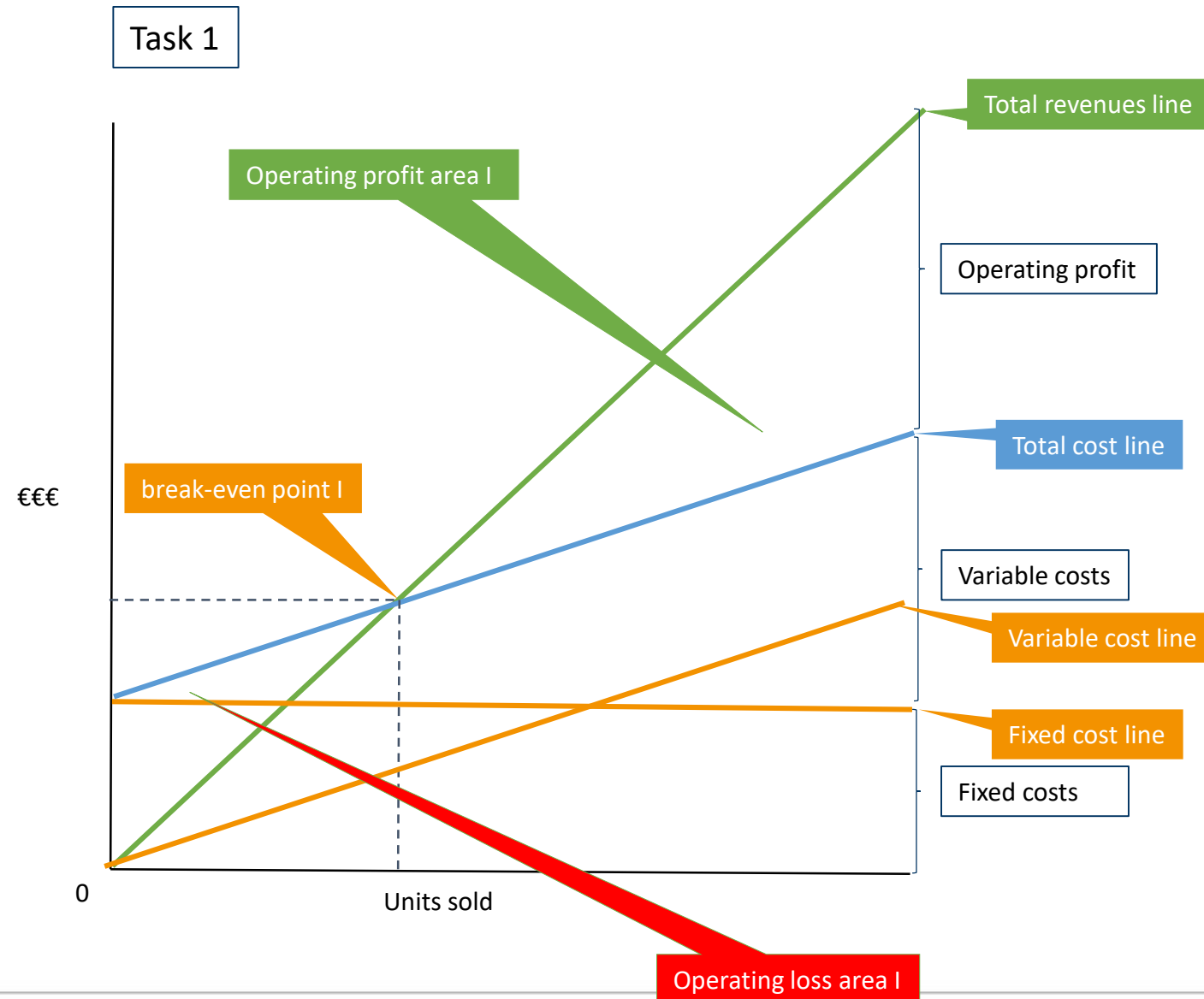
Y-axis: Euros

Task 2

Imagine the Buttercup Bakery decides to expand. On a second graph, redraw the CVP graph and show the theoretical effects of the expansion.

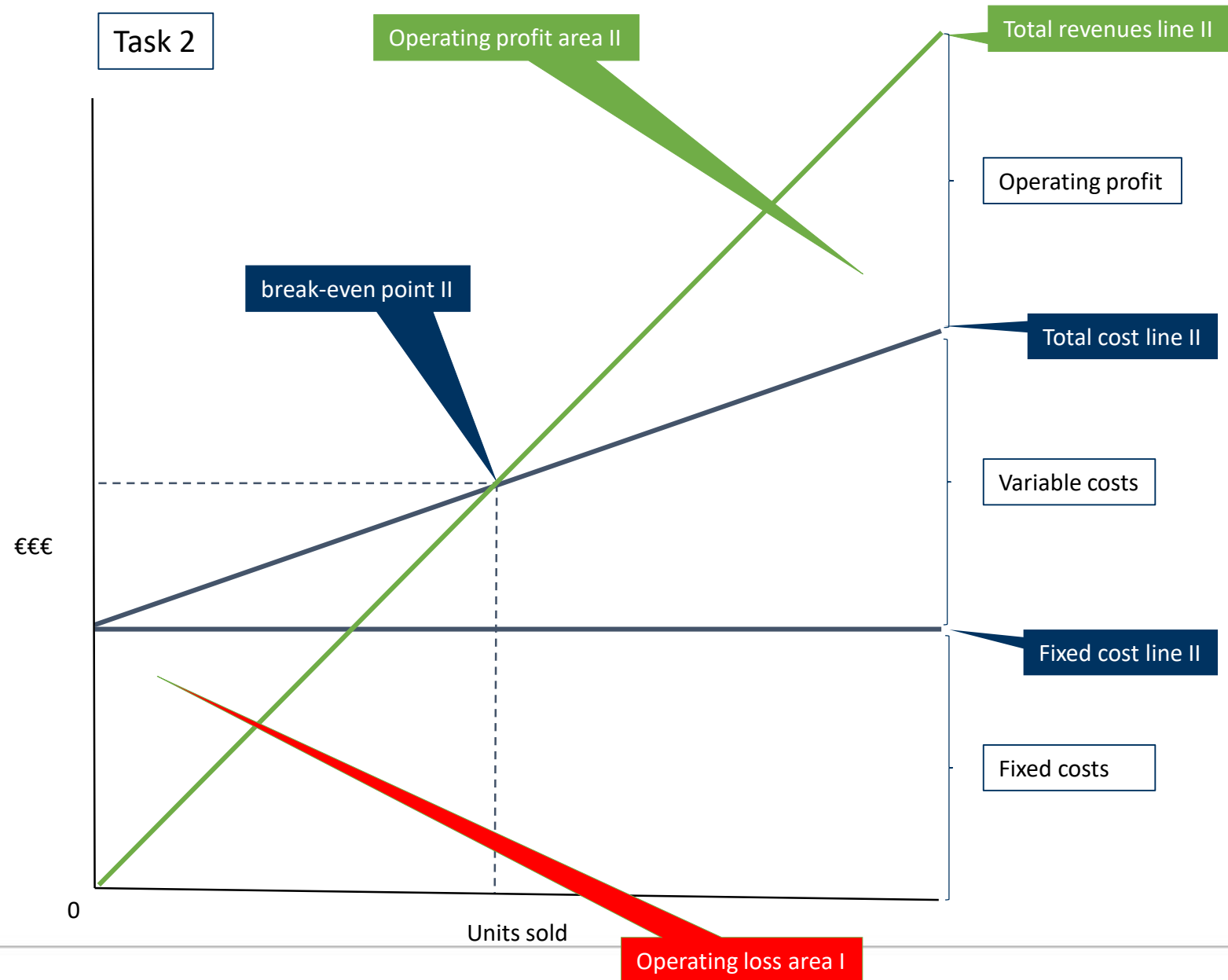
- Fixed cost line: Will it shift up or down? Why?
- Total cost line: Will the slope change? Why?
- Total revenue line: Will the slope change? Why?
- Breakeven point: Will it shift to the left or right? Why?

Exercise 3.2 - Solution



BHDR, 2023, p. 224, modified

Exercise 3.2 - Solution



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Exercise 10.11

Monteagudo-Playa SA is a take-away food store at a popular beach resort. Consuelo Herreros, owner of Monteagudo-Playa, is deciding how much refrigerator space to devote to four different drinks. Pertinent data on these four drinks are as follows:

	Cola	Lemonade	Punch	Natural orange juice
Selling price per case	€ 108,000.00	€ 115.20	€ 158.40	€230.40
Variable costs per case	€ 81.00	€ 91.20	€ 120.60	€181.20
Cases sold per metre of shelf-space per day	25	24	4	5

Consuelo has a maximum front shelf-space of 12 metres to devote to the four drinks. She wants a minimum of 1 metre and a maximum of 6 metres of front shelf-space for each drink.

Required:

- (1) What is the contribution margin per case of each type of drink?
- (2) A co-worker of Consuelo's recommends that she maximise the shelf-space devoted to those drinks with the highest contribution margin per case. Evaluate this recommendation.
- (3) What shelf-space allocation for the four drinks would you recommend for Monteagudo-Playa?

(BHDR, 2023, pp. 312-313)

Solution 10.11

Task 1)

	Cola	Lemonade	Punch	Natural orange juice
Selling price per case	€ 108.00	€ 115.20	€ 158.40	€ 230.40
Variable costs per case	€ 81.00	€ 91.20	€ 120.60	€ 181.20
Contribution margin	€ 27.00	€ 24.00	€ 37.80	€ 49.20

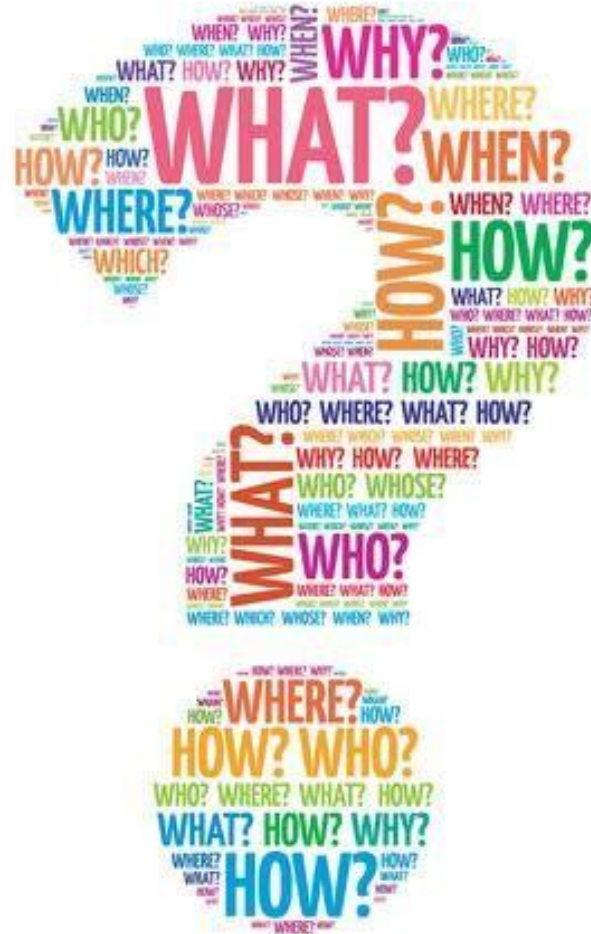
Task 2)

The CM per case is not informative about the maximum CM that can be achieved. Thus, we need to consider how much CM each product generates in relation to the existing restriction, i.e., metres as the constraining factor.

Task 3)

	Cola	Lemonade	Punch	Natural orange juice	
Daily CM per metre	€ 675.00	€ 576.00	€ 151.20	€ 246.00	
Metres of shelf space	6	4	1	1	12
Total CM per day	€ 4,050.00	€ 2,304.00	€ 151.20	€ 246.00	€ 6,751.20

Questions?



<https://www.pinterest.de/pin/799037158920259827/>

References

Bhimani, A., Horngren, C. T., Datar, S. M., & Rajan, M. V. (2023). *Management and Cost Accounting* (8th ed.). Pearson.